

Owl-Merge: Simple vs. Full Ontology Merge, GraphRAG Evaluation and a Multi-hop SPARQL Question-Answering Engine

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September 2026

1 Goal

Biomedical knowledge is spread over many databases (Hetionet, DRKG, PrimeKG, OptimusKG, OBO ontologies, DBpedia, NCBI Gene), each naming the same gene or drug with a different IRI. I received one pharmacological knowledge graph integrated two ways, as OWL files, and asked: *does the merge strategy change what an AI system can correctly answer?* The strategies follow Osman, Ben Yahia & Diallo, *Ontology integration: approaches and challenging issues*, Information Fusion (2021), DOI 10.1016/j.inffus.2021.01.007: **Simple Merge** $O_3 = O_1 \cup O_2 \cup O_A$ keeps equivalent entities separate and links them with `owl:equivalentClass` axioms carrying a confidence score; **Full Merge** $O_3 = O_1 \cup O_2$ fuses them into one entity that keeps one IRI and adds the others as labels. The paper argues both are semantically equivalent.

2 What I built, step by step

1. **Streaming OWL extractor** (`owl_extract.py`). The files are 954 MB and 659 MB, too big for in-memory libraries such as `rdflib`. A line-oriented regex parser streams RDF/XML into entity, edge and alignment tables in about 50 s.
2. **Structural analysis** with Python, `grep/awk`: entity and edge counts, equivalence links, class hierarchy, and a same-source ID check (a node holding two IDs from the same database, e.g. `Gene::675` and `Gene::5888`).
3. **GraphRAG evaluation with WeamRAG**, a hierarchical GraphRAG framework (GMM clustering, LLM community summaries, beam search). `build_weamrag_input.py` samples matched 3,000-node subgraphs from the same seed drugs and genes and converts them to WeamRAG's JSONL input; entity names carry their source (HLA-A [hetionet:3105]) so WeamRAG does not silently re-merge the Simple graph. Both indices were built locally (Llama 3.1 8B via Ollama, `nomic-embed-text`, `Milvus Lite`, `SQLite`).
4. **Question-answering test** (`compare_merge_rag.py`). The first version failed: embedding-only anchoring reached the target entities in 1 of 12 retrievals. I added entity-anchored seeding, a 10k-character context budget, reasoning-before-verdict prompts, two never-fused control pairs and a no-context control.
5. **Multi-hop SPARQL engine** (`kg_engine.py`). Both OWL files are bulk-loaded into embedded on-disk **Oxigraph** stores (`pyoxigraph`); every graph access is a SPARQL 1.1 query.
6. **Web application** (`web_app.py`, `web/index.html`). A dependency-free Python `http.server` backend with a validated JSON API and a plain HTML/CSS/JS page to ask a question against Simple, Full, or both side by side.

3 How the SPARQL engine answers a question

1. **Retrieval**: question phrases are matched to `rdfs:labels`, longest n-gram first, with a single-word fallback ("tamoxifen" → "tamoxifen citrate").
2. **Identity**: `owl:equivalentClass` links, plain or reified `owl:Axiom` with `pkg:confidenceScore`, are followed only above a threshold (default 0.9). This is how Simple Merge crosses databases.
3. **Reasoning**: two entities → bidirectional shortest-path search (≤ 3 relation hops); one entity → hop-by-hop expansion where type words ("drugs ... diseases") fix the target type per hop and relations are ranked by embedding similarity to the question (e.g. *indication* 0.58 followed, *contraindication* 0.50 skipped). The trace is shown.
4. **Answer**: Llama 3.1 8B answers only from numbered evidence paths and cites them ([P1]).
5. **Safety checks**: a *WARNING* for nodes fusing two same-source IDs and a *NOTE* when two named entities resolve to one node by label alone.

4 Results

Measure	Simple Merge	Full Merge
Declared entities	268,654	97,581
Edges	3,375,438	2,531,908
equivalentClass links (scored / unscored)	246,934 (208,688 / 38,246)	0
Classes with multiple parents	36.5%	35.2%
Entities holding ≥ 2 IDs from the same source	0	96
Relations in a matched 3,000-node sample	38,244	114,164
Triples in Oxigraph store (size)	5,343,202 (421 MB)	3,065,773 (238 MB)

- **Full Merge is 63.7% smaller but fuses different entities.** All 7 hand-checked same-source fusions are real errors: BRCA2+RAD51, atracurium+cisatracurium, estrone+estropipate, LGALS2+LGALS3, HLA-A+HLA-E, SYT11+SYT12, CERS1+GDF1.
- **Errors enter through identity chains.** LGALS2 was linked to Wikidata’s LGALS3 at 0.871; HLA-A reaches HLA-E in three links (weakest 0.701). The engine found $\text{BRCA2} \equiv \text{dbpedia:RAD51 (unscored)} \equiv \text{RAD51 (0.72)}$.
- **GraphRAG passes the error to the LLM.** With no context the model judged all 7 pairs *different*. With Full Merge context it answered *same* for 4 of 5 fused pairs, quoting the fused node as evidence; with Simple Merge context it said *same* once. Small test (7 pairs, one 8B model), so suggestive, not conclusive.
- **The engine shows the difference directly.** “Are BRCA2 and RAD51 the same gene?” returns **BRCA2 -protein protein-> RAD51** (they interact) on Simple Merge, but only a fused node plus a WARNING on Full Merge. “How is BRCA2 related to tamoxifen?” crosses three databases via 0.95 identity links on Simple Merge. Queries take seconds without the LLM.

5 Technology used

Data & standards	OWL, RDF/XML, SPARQL 1.1, reified owl:Axiom alignments
Triple store	Oxigraph (pyoxigraph 0.5.11), embedded, on-disk
GraphRAG	WeamRAG, Milvus Lite, SQLite, UMAP/GMM clustering
Models (local)	Ollama: Llama 3.1 8B (answers), nomic-embed-text (relation ranking); NVIDIA TITAN RTX
Backend & frontend	Python 3 standard library (http.server), HTML/CSS/JavaScript, no build step
Tooling	Git/GitHub, Playwright (UI testing), VS Code Remote-SSH

6 How to run

```
python3 -m venv .venv && .venv/bin/pip install pyoxigraph==0.5.11
ollama pull llama3.1:8b && ollama pull nomic-embed-text
.venv/bin/python kg_engine.py load simple && .venv/bin/python kg_engine.py load full
.venv/bin/python kg_engine.py ask simple "How is BRCA2 related to tamoxifen?"
.venv/bin/python web_app.py # open http://127.0.0.1:8780
```

7 Limitations and next steps

Only the first two named entities in a question are connected; name matching is exact or single-word, so misspellings (e.g. “metformonin”) do not match; target types come from IRI patterns; the QA test is small. The two OWL files were not built from exactly identical inputs, so some differences reflect inputs rather than strategy. Next: review the remaining 89 same-source fusions, add fuzzy entity matching, and evaluate on a larger question set with a stronger model.